

Local Anaesthetics

Definition

Local anesthetics reversibly block impulse conduction along nerve axons & other excitable membrane that utilize Na^+ channels for an action potential generation.

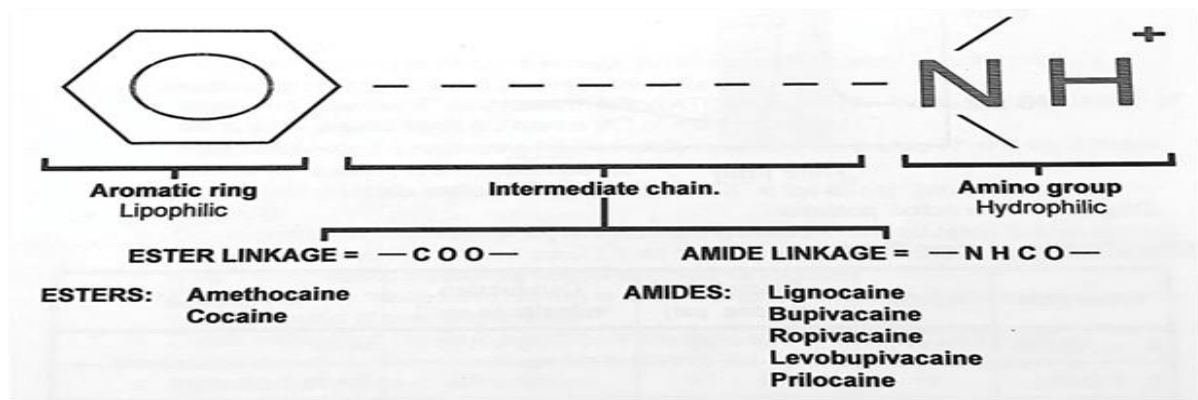
It blocks pain sensation (nociception) from specific areas of the body.

Characteristics of ideal local anesthetic

- 1- Rapid onset.
- 2- Long duration of action.
- 3- Reversible & selective blockade of sensory nerves without motor blockade.
- 4- Minimal local tissue irritation & no systemic toxicities (cardiac & CNS).

Chemistry

All local anesthetics contain 3 structural components:



1- Aromatic ring: lipophilic group for membrane penetration

2- Connecting group: This is either

-an ester (e.g., cocaine, procaine, tetracaine and benzocaine)

-or an amide (e.g. lidocaine, bupivacaine, mepivacaine and ropivacaine).

- *The nature of the linking group determines some of the pharmacological properties of these agents. For example, local anesthetics with an ester link are hydrolyzed readily by plasma esterase in both the plasma mainly and liver as **procaine**.*
- *The liver is the chief site of the destruction of the amide type of local anesthetics as e.g. **lidocaine**.*

3- Ionizable amino group: for channel blockade.

Effect of pH

- Local anesthetics are weak bases and are usually made available clinically as salts to increase solubility and stability. In the body, they exist either as the uncharged base or as a cation.
- Local anesthetic activity is increased at alkaline pH (i.e. when the proportion of ionized molecules is low) and reduced at acid pH. This is because the compound needs to penetrate the nerve sheath and the axon membrane to reach the inner end of the sodium channel (where the local anesthetic-binding site resides).
- Because the ionized form is not membrane-permeant, penetration is very poor at acid pH.

This pH dependence can be clinically important, because inflamed tissues are often acidic and thus somewhat resistant to local anesthetic agents.

Mechanism of action:

Local anesthetics prevent both the generation and conduction of the nerve impulse. Their main site is the cell membrane and they induce changes in the permeability of the membrane.

1. They block conduction by interfering with the transient increase in the permeability of the membrane to sodium ions through direct interaction with voltage sensitive Na channels.

2. They reduce the permeability of the resting nerve to potassium as well as Na^+ ions. Thus, they block conduction without depolarizing the nerve i.e. they stabilize the membrane action potential.
3. They compete with calcium at some receptor site that controls the permeability of the membrane.

Mechanism of action cont...

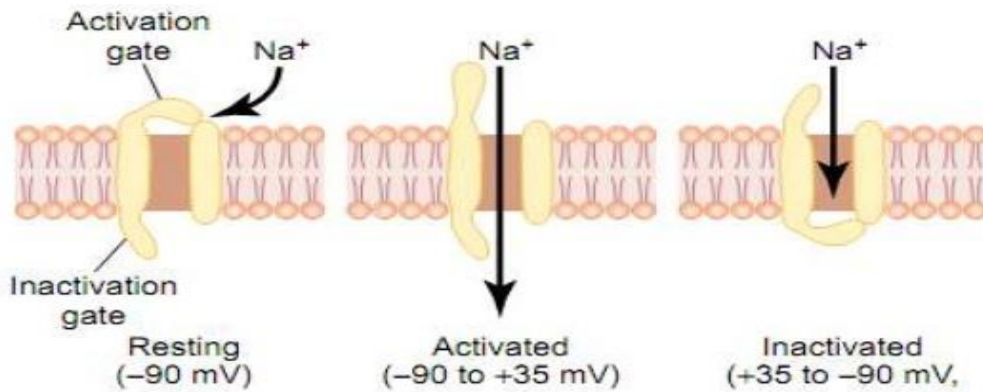


Figure: Sodium channel

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Pharmacokinetics

- Local anesthetics are generally not absorbed from intact skin. When applied to mucous membranes, the absorption varies with the mucous surface, e.g. absorption is more rapid from the trachea than from the pharynx while it is little through the urinary bladder.
- When used by infiltration, absorption can be retarded by using vasoconstrictor agents along with the drug.
- Most of the common local anesthetics are esters and their metabolic degradation by hydrolysis by esterase in both the plasma mainly and liver as **procaine**.
- The liver seems to be the chief site of the destruction of the amide type of local anesthetics as e.g. lidocaine.

Differential nerve sensitivity:

- Small non-myelinated nerve fibers were more susceptible to the action of local anesthetics than are larger myelinated fibers. Unfortunately, they are the slowest to recover.
- The sensation of pain is usually the first modality to disappear and is followed in turn by the sensations of cold, warmth, touch and deep pressure.
- Motor axons (large diameter), are relatively resistant.

Structure- Activity Characteristics

- Smaller & more lipophilic members have a faster rate of interaction with Na⁺ channels.
- Potency is positively correlated with lipid solubility. Tetracaine, bupivacaine, & ropivacaine are more lipid soluble than lidocaine, procaine, & mepivacaine so they are more potent & have longer duration of action.
- Long acting (bupivacaine) also bind more extensively to plasma proteins & can be displaced by other protein-bound drugs.

Prolongation of Action by vasoconstrictors

- In clinical practice, a vasoconstrictor, is often added to local anesthetics. The vasoconstrictor performs the following services by decreasing the rate of absorption:
 1. It localizes the anesthetic at the desired site so increases the duration of action.
 2. Reduces the systemic toxicity by keeping the space between the rate of absorption and the rate of destruction in the body.
 3. Vasoconstriction produced decreases bleeding and provides a bloodless field during the operation.

4. Epinephrine incorporated into a spinal anesthetic may not only contribute to prolongation of the local anesthetic effect but also exert a direct analgesic effect mediated by postsynaptic α_2 adrenoceptors within the spinal cord.
- Epinephrine is widely used in concentration of 1: 200,000 to 50,000 in solution of local anesthetics.
 - Phenylephrine(0.05%) can be used as an alternative to epinephrine.
 - Sympathomimetic amines **increase oxygen consumption** of the tissues and together with the **vasoconstriction** will lead, to hypoxia and local tissue damage → delay in wound healing, tissue edema or necrosis after local anesthesia.
 - **Cocaine** inhibits the neuronal membrane transporters for catecholamines, thereby cocaine by itself is a vasoconstrictor by potentiating the effect of NE at α -adrenergic receptors in the vasculature, resulting in vasoconstriction, also **ropivacaine** and **bupivacaine** also cause vasoconstriction-- so the addition of a vasoconstrictor agent to them is not needed

Toxicity

- Local anesthetics, not only interfere with conduction in the peripheral nervous system, but also with the function of all organs in which conduction or transmission of impulses occurs. Thus, they have important effects on the **CNS**, the **autonomic ganglia**, the **neuromuscular junction**, and **all forms of muscle**.
- Local anesthetic toxicity derives from two distinct processes: (1) systemic absorption of the local anesthetic from the site of administration; and (2) neurotoxicity resulting from local effects produced by direct contact with neural elements.

1- CNS

- CNS stimulation, producing restlessness and tremor that may progress to clonic convulsions.

- Central stimulation is followed by depression; death usually is caused by respiratory failure.
- ***Cocaine*** has a particularly prominent effect on mood and behavior ***and is capable of producing addiction.***

2- CVS

- It decreases the electrical excitability, conduction rate, and force of contraction.
- Unwanted cardiovascular effects usually are seen only after high systemic concentrations are attained and effects on the CNS are produced.
- Rarely, lower doses of some local anesthetics will cause cardiovascular collapse and death.
- ***All Local anesthetics cause vasodilation Except cocaine is a vasoconstrictor by itself.***
- ***Bupivacaine and levobupivacaine cause vasodilatation at clinical doses, but lower doses appear to cause vasoconstriction.***
- ***Bupivacaine produce serious arrhythmia and tachycardia.***

3- Smooth Muscle

- Local anesthetics depress contractions of gastrointestinal, vascular and bronchial smooth muscle. Uterine contractions may be depressed directly during intrapartum regional anesthesia.

4- Neuromuscular junction

Local anesthetics in large doses produce neuromuscular blockade by an action that is different from the local anesthetic action on the nerve.

5- Hypersensitivity to Local Anesthetics

Rare individuals are hypersensitive to local anesthetics. The reaction may manifest itself as an allergic dermatitis, or a typical asthmatic attack. ***Hypersensitivity seems to occur more frequently with local anesthetics of the ester type***

Classification

(1) Natural local anesthetics as cocaine.

(2) Synthetic local anesthetics; these are further classified according to their physical properties into:

(a) ***Soluble*** salts as procaine, lidocaine, tetracaine, dibucaine.

-They are suitable for infiltration or injection and sometimes also for topical application.

(b) ***Slightly soluble*** as benzocaine (ethyl aminobenzoate).

-These are esters of Para aminobenzoic acid but they lack tertiary amino group. Their limited solubility in water renders them unsuitable for injection purposes.

Techniques for local anesthesia

(1) Surface anesthesia

- The anesthetic is applied to the surface of the skin and mucous membranes to relieve pain due to ulcers, fissures, burns...etc. and ophthalmologic, rhinolaryngological and urological practice.

(2) Infiltration anesthesia

- The anesthetic is injected into the skin or sc. tissues to anesthetize nerve ending by their direct exposure to the drug.

- ***Ring block***, a form of infiltration involves infiltration of the anesthetic subcutaneously in a circular manner around the field of the operation, thus effectively blocking sensor nerves supplying the skin over the operative field.

(3) Block anesthesia

- They range from block of a single nerve through various plexus blocks to epidural and spinal anesthesia. It diffuses from the site of injection into the nerve, where it acts.

(4) Spinal anesthesia

- Spinal anesthesia (subarachnoid or intrathecal block) is a type of extensive block anesthesia in which the anesthetic is injected into the subarachnoid space between the second and fifth lumbar levels.
- Spinal anesthesia may be accompanied by a profound drop in blood pressure due to dilatation of the blood vessels in the viscera and lower extremities as a result of paralysis of sympathetic outflow from that portion of the spinal cord exposed to the anesthetic agent.

Neurological complications in the form of:

- Headache. - Septic meningitis.
- Numbness in the area supplied by the lumbosacral roots.
- Traumatic nerve injury may follow lumbar puncture.

Properties of local anesthetics

Lidocaine

- lidocaine is packaged in two major formulations: a) as monotherapy and b) in combination with epinephrine for local anesthetic use to prolong the duration of local anesthesia
- Blocks voltage sensitive Na channels in nerves and cardiac tissue
- It is a local anesthetic (amide) and class 1B antiarrhythmic

Indications:

- Production of local or regional anesthesia
- Treatment of ventricular cardiac arrhythmias (esp. post-MI)
- Not effective against supraventricular arrhythmias (atrial or AV node)

Side Effects: (related to overdose)

- light-headedness, tinnitus, metallic taste, blurred vision, numbness, twitching, convulsions, hypotension

- twitching, convulsions - due to Na channel block in CNS inhibitory centers
(more likely to occur at high doses e.g. > 9ug/ml)

Black box warning

- Do Not Use Lidocaine to Treat Teething Pain in Children.
- Rubbing topical pain relievers on gums is not useful or effective because they wash out of the baby's mouth within minutes.
- when too much is swallowed, it can result in seizures, severe brain injury, and cardiac toxicity.

Pharmacokinetics

- Hepatic metabolism, half life 1.5-2 hrs
- Patients with CHF or hepatic disease will have higher plasma levels (this can result in CNS toxicity)

Bupivacaine

- Parenteral only
- Packaged in combination with epinephrine for local anesthetic use to prolong the duration of local anesthesia

Side Effects: (related to overdose)

- light-headedness, tinnitus, metallic taste, blurred vision, numbness, twitching, convulsions.
- *Bupivacaine produce serious arrhythmia and tachycardia* (potentially lethal)

Pharmacokinetics:

- longer duration of action than most other local anesthetics (e.g. vs. lidocaine)
- 16 times more potent than lidocaine

Ropivacaine (long duration of action similar to bupivacaine)

Mepivacaine

Pharmacokinetics:

- Mepivacaine (medium duration of action than most other local anesthetics (e.g. vs. lidocaine))
- Widely used,
- Used IV in ventricular arrhythmia.
- *It is highly toxic to neonates and not used in obstetrical anesthesia*

Cocaine

- Natural
- It is vasoconstrictor by itself
- Produce Addiction
- Used in surface anesthesia only

Prilocaine

- Has medium onset and duration
- Can be used without a vasoconstrictor.
- Produce methemoglobinemia, so not used for obstetric operation

Tetracaine

- Has very slow onset and long duration of action
- Used in spinal & corneal anesthesia
- Side effects as lidocaine